

Aaromal A.

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Electronics Engineering · AI/ML Research

EDUCATION

Central Board of Secondary Education (CBSE)

Kerala, 2022 – 2024

Higher Secondary Education, Mathematics and Computer Science

Grade: 94.6%

Coursework: Python Programming, Database Management Systems (DBMS), Data Structures and Algorithms (DSA), Computer Networking, Statistics & Probability, Coordinate Geometry, Calculus.

Activities: Kerala State Junior Meet Athlete, District Junior Athletic Meet Medalist, School Cadre Group Leader.

Cochin University of Science & Technology (CUSAT)

Kerala, 2024 – 2028

B.Tech in Electronics & Communication Engineering

CGPA: 8.67 (S3)

Major Coursework: Analog Integrated Circuits, Digital System Design, Signals and Systems, Network Theory, Microprocessors & Microcontrollers (8086, 8051), C programming & Object-Oriented Programming (C++), Linear Algebra & Transform Techniques, Numerical & Statistical Methods, Calculus.

RESEARCH & PRESENTATIONS

Ecometachip Mini: Low-Cost, Eco-friendly, Dielectric Chemical Identifier

New Delhi

Lecture Presentation, IEEE APSCON 2026 (Student Research Forum)

23-25 Feb 2026

- Designed a eco-friendly dielectric sensing system for preliminary identification of common liquids (water, ethanol, acetone, petrol) based on relative permittivity variation in a biodegradable coir-rubber composite as the primary sensing element.
- Developed a 100 MHz Hartley oscillator-based readout circuit with RF pickup for magnetic field modulation proportional to variation in rel. permittivity of material with different liquids and precision rectifier for rectification in following stages.
- Implemented LED-based RGB visual classification to validate output voltage variations (low cost).
- Simulated ideal behavior in **LTspice** by varying coupling constants and analyzed the sensitivity and repeatability of the sensor under variable environmental conditions (Humidity/Temp) using **Python (Pandas/Seaborn)**.
- Quantified environmental drift coefficients ($mV/\%RH$ and $mV/^{\circ}C$) to characterize sensor robustness and output consistency across variable ambient conditions.
- Prepared research paper using $\text{\LaTeX}$ which got selected for lecture presentation under [Student Research Forum, IEEE APSCON'26 \(IEEE Applied Sensing Conference\)](#) by [IEEE Sensors Council](#) held at New Delhi, 23-25 Feb, 2026. ([ResearchGate presentation DOI](#))

Attention-Regularized Vision Transformer for Cloth-Agnostic Person Re-Identification.

Kottayam

Ongoing Research, Institute Internship Programme, IIIT Kottayam

May 2026 – Present

- Researching cloth-agnostic person re-identification (**CC-ReID**, **PU-ReID**) under [Dr. Sridhar Raj S](#), begun as a summer research internship under the IIITK Institute Internship Programme.
- Writing, training, and evaluating **Vision Transformer (ViT-Base, PyTorch)** models with regularized attention to improve robustness to clothing changes.
- Conducted multiple ablation studies, achieving improvements over current benchmark models on the task. Writing a journal paper out of it and in process on publishing on a journal.

FEATURED PROJECTS

github.com/aaromalonline

Liberate:ATCS - Adaptive Typing & Control System

Feb 2025 - Sep 2025

ADXL345, TCRT5000 IR, ESP32, C++, Python, Arduino IDE

- Implimented muscle-twitch decoding interface using **ADXL345 accelerometer** (also **TCTRT5000 IR sensor**) as input modules to record muscle twitches as 3-axis accelerations, which are encoded into binary clicks using baseline calibration and filtering algorithms.

- Data is sent serially/wirelessly via ESP32 to the **Python application** using I2C, where the user can control a keyboard interface using the decoded twitch clicks.
- The **PyQt5 interface** provides a moving highlight bar which spans across the rows of a adaptive keyboard interface (incl QWERTY, SOS, Mobility, Speak Keys), allowing selection of rows and keys through muscle twitches.
- The decoded twitch signals can be used for various applications incl communication, mobility (used to move wheelchair in addition to a motor-contolller) etc, aiming to assist people with ALS or paralysis.
- Received participation recognition at the **2025 IEEE SSCS Arduino Content** and **STRIDE Assitive Designation 2025** (Social Technology & Research for Inclusive Design Excellence) by **K-DISC** (Kerala Development and Innovation Strategic Council) documenting **video presentation** and demonstration of the prototype.

CardioKey (ECGID) - Analog ECG Acquisition & Biometrics

Feb 2026 - Present

AD8232 ECG & ADS1115 ADC, Arduino UNO, Python, Numpy & Scipy, Matplotlib, Scikit-learn, Tkinter

- Engineered an **lead-1 analog front-end** using the AD620 instrumentation amplifier and filter stages (Right-leg drive, Notch, butterworth) to acquire low-amplitude (1mV) heart signals amplified 1000x in V range using amplifier stages.
- Digitally sampled the analog out in 300-500 Hz, 16 bit precision using 16 bit ADC(ADS1115) with DC offset circuit (Clamper) and Arduino UNO as a data pipe to the processing computer (**client-side architecture**).
- Digital signal undergoes segmentation, fragment selection extracting P-QRS-T feature space, then Principal Component Analysis (PCA) to compress 250-point cardiac windows into 30-point biometric "signatures". (**Feature space formation & dimensionality reduction**)
- Achieved a 96% identification accuracy using Linear Discriminant Analysis (LDA) with Nearest Mean and majority vote classifiers on 30pt pca signature of 45 subjects (in ages 18-22) extracting 10 beats (P-QRS-T fragment) each, divided into 70-30 to train and predict, data acquired from public database. (**Classification & Identification**)
- Deployed a functional client-side testing environment featuring high-frequency Arduino sampling firmware and a custom **Tkinter-based interface** for real-time ECG recording and prediction. Architecting an edge-processing pipeline to migrate signal analysis from the client-side to Raspberry Pi/ESP32 platforms using **TinyML** for low-latency, on-device inference.
- Featured inherent "**Liveness Detection**" to prevent biometric spoofing common in fingerprint or face ID systems and hence proved uniqueness of normalised P-QRS-T ecg fragment across people, age and heart rate variability.

Wake - On-Device AI Memory for Android

Nov 2026

Kotlin, Jetpack Compose (Material 3), Gradle, Room (SQLite/FTS4), MediaPipe TextEmbedder, LiteRT-LM, Gemini API

- Built a local-first, on-device AI agent that captures useful text from Android notifications and the visible screen via an **Accessibility Service** and **Notification Listener Service**, storing it in a single **MemoryEvent** table – no screenshots or pixels are ever captured – and answers user queries about what was seen or received.
- Implemented an **agentic architecture**: **AgentReasoner** decides when to act, **CommitmentDetector** identifies user promises/intents from captured text, **ActionExecutor** executes actions, and **AgentEngine** orchestrates the perception → reasoning → action loop.
- Used **MediaPipe TextEmbedder** for on-device embeddings and **LiteRT-LM** for on-device **Gemma4:E2B** inference, with the Gemini API as an optional cloud LLM, in a single-module, no-DI, privacy-first architecture.
- Built at the **Google DeepMind × Cerebral Valley** hackathon (Bengaluru) using the latest Google AI stack (Gemini 3.5 Flash, Gemini Audio, GenMedia – Omni, Nano Banana 2 Lite – Interactions API/Antigravity, Gemma4); selected as one of the top 15-20 teams from 4,000+ applicants (~350 shortlisted).

Digital Logic Gate IC Tester

2026 – Present

AT89S52 (8051), Embedded C (SDCC/Keil C51), Arduino UNO, 16x2 LCD

- Designing and building a low-cost microcontroller-based tester for verifying the functional operation of selected 74-series combinational logic ICs (**74LS00 NAND, 74LS02 NOR, 74LS04 NOT**), using an **AT89S52** 8051-family microcontroller as the central controller and a 14-pin ZIF socket to mount the Device Under Test (DUT).
- Push buttons provide user input for IC/test selection, while a **16x2 LCD** displays the selected IC and test status.
- Firmware applies the required input combinations, reads the corresponding outputs, and compares them against truth-table values stored in firmware to indicate pass/fail of the DUT.
- Validated the circuit through simulation before implementing on perfboard; firmware developed in **Embedded C** (SDCC or Keil C51) with an Arduino UNO used as the programming interface for the AT89S52.

PassCrypt - Password Manager

2022

Python, SQLite3

A local password manager built with **Python** and **SQLite** for secure credential storage, using **SHA-256** hashing for user authentication and **AES encryption** (via `cryptography.fernet`) to keep stored passwords secure. Includes an advanced search/filter interface and a data-portability feature for exporting/importing the entire credential database.

Truevote - Digital Voting System

May 2025

C++, Qt, SHA-256 Hashing, CMake

A cross-platform digital voting application built with **C++** and **Qt**, featuring an OOP-based admin/voter panel architecture for poll creation, voter registration, and authentication, with votes recorded and automatically tallied from CSV files. Uses **SHA-256** hashing for authentication and a **CMake**-driven build-and-run workflow for Linux.

EXPERIENCE

Technical Lead (R&D AI/Robotics)

Jun 2026 – Present

Team SingularIT, CUSAT

Kochi, India

- Leading R&D on autonomous systems as Technical Lead for **Team SingularIT**, a student team preparing an autonomous entry for **A2RL** autonomous racing.
- Architected and developed the **perception subsystem** for the autonomous racing platform, running on an **NVIDIA Jetson** paired with an **STM32H743** ARM flight controller.
- Designed the perception stack to rely solely on **IMU** and **vision** (no external localization aids), fusing visual and inertial data for real-time state estimation.

AI Research Intern – Person Re-Identification

May 2026 – Present

Indian Institute of Information Technology Kottayam (IIITK)

Kottayam, India

- Conducting AI research under **Dr. Sridhar Raj S** on cloth-agnostic person re-identification (**CC-ReID, PU-ReID**), begun as a summer research internship under the Institute Internship Programme.
- Working on regularizing the attention of **Vision Transformers (ViT-Base, PyTorch)** to improve robustness to clothing changes.
- Wrote, trained, and evaluated vision models from scratch, running ablation studies that achieved improvements over current benchmark models. ([Google Scholar](#))

Software Intern (*works*)

Mar 2025 – Jul 2025

Horizon Mars Rover Team, CUSAT

Kochi, India

- Dual-booted **Ubuntu 22.04** by manual partitioning and setup **ros-humble-desktop-full** apt package.
- Architected node communication for a functional Mars rover using **ROS 2 Humble** and **Arduino framework**, wrote simple subscriber & publisher nodes and proximity alert system using them in **Python & C++**.

- Performed circuit simulation using **TinkerCAD** and **Arduino** to program robotic arm joints using servo-motors.
- Wrote technical reports of the rover for upcoming IROC and ERC competitions using **L^AT_EX**
- Track record for qualifying AIR 26 on IRC26' and global rank 18 on ERC24' rover challenges including the **European Rover Challenge (ERC)** and **IROC by ISRO**, India.

Startup Founder | Founder Next Door (8-week mentorship program)
IEDC CUSAT, CITIC

Oct 2025 – Dec 2025
Kochi, India

Selected from a campus-wide pool of proposals for an intensive 8-week incubator on market validation and MVP architecture. Designed the **system architecture** and business logic for **Dozyo**, a hyperlocal platform for student-led gigs, and developed a comprehensive **business plan** using Telegram community, Airtable, and escrow services to replace informal job-sharing with structured digital workflows. Collaborated weekly with industry founders on ideation, branding & marketing, and technical scalability, concluding with an MVP presentation.

VOLUNTEERING

Active Member
FOSS CUSAT | Community

Mar 2025 – Present
Kochi, India

Spearheaded the **Linux Installation Party** under the **Endof10 Campaign**, onboarding 20+ students to open-source operating systems with hands-on guidance on **manual disk partitioning**, Secure Boot configuration, and shell workflows across **Debian, Arch, Fedora, Ubuntu, and Pop!_OS**, using deployment tools like **Ventoy** and **Balena Etcher**. Represented FOSS CUSAT at **Kochi FOSS Meetups** and **IndiaFOSS 2025** (Bengaluru).

Community Volunteer
Tinkerspace, TinkerHub Foundation

2026 – Present
Kochi, India

Host and actively engage in discussions at **Tinkerspace**, a local makerspace by TinkerHub, running **AI discussion sessions** and a **paper club**, mentoring at in-space hackathons like **Useless Projects**, and connecting with founders and fellow makers on their projects.

TECHNICAL SKILLS & CERTIFICATIONS

- **Cohere Labs Open Science Community ML Summer School '26 (Online):**
 - Attended sessions led by research scientists, PhD researchers, and mathematicians from **Meta Superintelligence Labs**, **Google DeepMind**, **Cohere Labs**, and **Liquid AI** on fundamental ML math, computer vision, and reinforcement learning.
 - Covered vector calculus and linear algebra foundations of ML algorithms, the evolution of computer vision from classical techniques to transformer-based and vision-language architectures, and value-based/policy-based reinforcement learning.
- **MATLAB Academy (Online Summer Internship):** *Jun 2025*
 - **MATLAB Proficiency:** Core programming, algorithm development, and signal processing.
 - **Data Analysis & Processing:** Visualization and AI/Machine Learning modules.
 - **MATLAB for Simulink Users:** System modeling, circuit simulation, and control logic.
- **Languages:** Proficient in **Python** (NumPy, SciPy, Matplotlib, Seaborn, Scikit-learn, Pandas, OpenCV, Requests, Selenium, SQLite), **C/C++**, **x86 asm**, **HTML/CSS**, **JS**, **Bash** (basic scripting), **SQL**, **L^AT_EX** and **Verilog**.
- **Tools & Frameworks:** **ROS 2 Humble** (Basic Nodes), **Qt**, **Arduino Framework**, **LTspice**, **Xilinx Vivado**, **Proteus**, **PyTorch**, **Kotlin/Jetpack Compose**, and **MATLAB & Simulink** (Basics), **MySQL**.
- **Linux & DevOps:** **Git** workflow; Daily-driving **Linux** for >5 years across **Debian** (current), **Ubuntu**, **Fedora**, and **Arch**; experienced in VM management, manual SSD/HDD partitioning, **Ventoy**, and **Balena Etcher**.